Knowledge Compilation Modulo Theories for Weighted Model Integration

M.Sc. in Computer Science

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- Background & State of the Art
- Novel Contributions
  - KCMT for Volume Computation
  - KCMT for WMI
- Conclusions

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$\mathcal{LRA}$ formula:

$$\varphi \stackrel{\text{def}}{=} (A \vee (x > 10)) \wedge (\neg A \vee (x < 5)) \wedge ((x > 10) \vee (x < 5))$$

$\mathcal{LRA}$ formula:

$$\varphi \stackrel{\text{def}}{=} (\overset{\mathbb{B}}{A} \vee (x > 10)) \wedge (\neg \overset{\mathbb{B}}{A} \vee (x < 5)) \wedge ((x > 10) \vee (x < 5))$$

$\mathcal{LRA}$ formula:

$$\varphi \stackrel{\text{def}}{=} (\overset{\mathbb{B}}{A} \vee (\overset{\mathcal{LRA}}{x > 10})) \wedge (\overset{\mathbb{B}}{\neg A} \vee (\overset{\mathcal{LRA}}{x < 5})) \wedge ((\overset{\mathcal{LRA}}{x > 10}) \vee (\overset{\mathcal{LRA}}{x < 5}))$$

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Truth Assignments:

$$\mu_1 = \begin{cases} A & \mapsto \top \\ (x > 10) & \mapsto \top \\ (x < 5) & \mapsto \top \end{cases} \qquad \mu_2 = \begin{cases} A & \mapsto \top \\ (x > 10) & \mapsto \bot \\ (x < 5) & \mapsto \top \end{cases}$$

$\mathcal{LRA}$ formula:

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$$\mu_2 = \begin{cases} A & \mapsto \top \\ (x > 10) & \mapsto \bot \\ (x < 5) & \mapsto \top \end{cases}$$

- $\varphi_{\left[\mu_1\right]} \equiv_{\mathbb{B}} \varphi\left[\psi / \mu_1(\psi)\right] \equiv_{\mathbb{B}} \top$
- $\varphi_{\left[\mu_2\right]} \equiv_{\mathbb{B}} \varphi\left[\psi / \mu_2(\psi)\right] \equiv_{\mathbb{B}} \top$

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Truth Assignments:

$$\mu_1 = \begin{cases} A & \mapsto \top \\ (x > 10) & \mapsto \top \\ (x < 5) & \mapsto \top \end{cases}$$

- $\varphi_{\left[\mu_1\right]} \equiv_{\mathbb{B}} \varphi\left[\psi / \mu_1(\psi)\right] \equiv_{\mathbb{B}} \top$
- $\mathcal{T}$-inconsistent: $(x > 10), (x < 5)$

$$\mu_2 = \begin{cases} A & \mapsto \top \\ (x > 10) & \mapsto \bot \\ (x < 5) & \mapsto \top \end{cases}$$

- $\varphi_{\left[\mu_2\right]} \equiv_{\mathbb{B}} \varphi\left[\psi / \mu_2(\psi)\right] \equiv_{\mathbb{B}} \top$
- $\mathcal{T}$-consistent

$$\psi \stackrel{\text{def}}{=} (\neg A_1 \wedge A_2) \vee (\neg A_2 \vee A_3)$$

[1] A. Darwiche and P. Marquis. "A Knowledge Compilation Map". In: *Journal of AI Research* (2002)

Knowledge Compilation

Background & State of the Art

$$\psi \stackrel{\text{def}}{=} (\neg A_1 \wedge A_2) \vee (\neg A_2 \vee A_3)$$

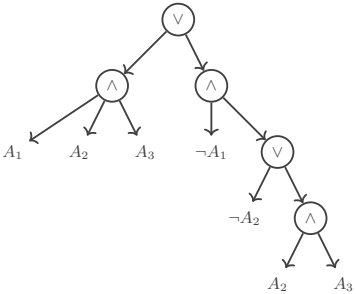

Figure: $\texttt{d-DNNF}\big(\psi\big)$.

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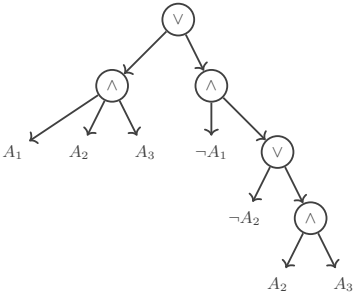

Figure: d-DNNF(ψ).

Operation	NNF	CNF	d-DNNF	SDD
Queries				
model counting	○	○	✓	✓
model enumeration	○	○	✓	✓
Transformations				
conditioning	✓	✓	✓	●
forgetting	○	○	○	●

✓ "satisfies" ● "does not satisfy" ○ "does not satisfy unless P = NP"

Table: Subset of the NNF languages and their corresponding polytime queries and transformations[1].

[1] A. Darwiche and P. Marquis. "A Knowledge Compilation Map". In: *Journal of AI Research* (2002)

Knowledge Compilation Modulo Theories

Background & State of the Art

$$\varphi \stackrel{\text{def}}{=} (A \vee (x > 10)) \wedge (\neg A \vee (x < 5)) \wedge ((x > 10) \vee (x < 5))$$

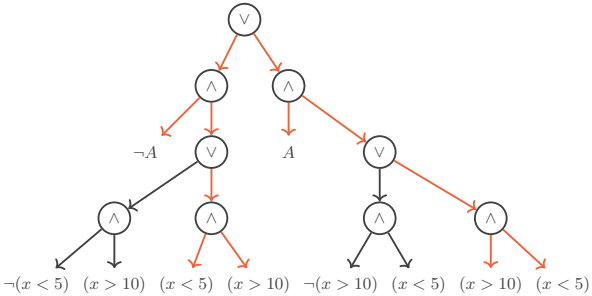

Figure: $\texttt{d-DNNF}\big(\varphi\big)$.

[1] M. Michelutti, G. Masina, G. Spallitta, and R. Sebastiani. "Canonical Decision Diagrams Modulo Theories". In: *ECAI*. 2024

Knowledge Compilation Modulo Theories

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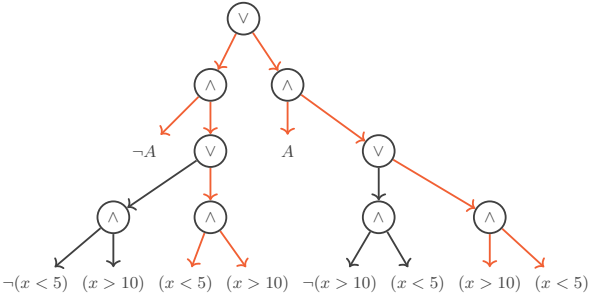

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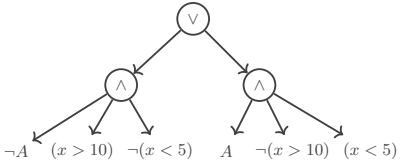
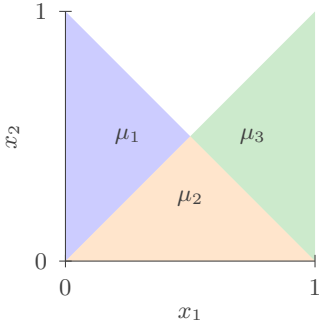


Figure: $\texttt{d-DNNF}\Big(\varphi \wedge \underbrace{\big((\neg(x > 10) \vee \neg(x < 5))\big)}_{\mathcal{T}\text{-lemmas}(\varphi)}\Big)$[1].

[1] M. Michelutti, G. Masina, G. Spallitta, and R. Sebastiani. "Canonical Decision Diagrams Modulo Theories". In: *ECAI*. 2024

Volume Computation

Background & State of the Art



$$\varphi \stackrel{\text{def}}{=} (0 \leq x_1) \wedge (0 \leq x_2) \wedge ((x_1 + x_2 \leq 1) \vee ((x_1 \geq x_2) \wedge (x_1 \leq 1)))$$

$$\mu_1 = \left\{ (0 \leq x_1), (0 \leq x_2), (x_1 \leq 1), \quad (x_1 + x_2 \leq 1), \neg(x_1 \geq x_2) \right\}$$

$$\mu_2 = \left\{ (0 \leq x_1), (0 \leq x_2), (x_1 \leq 1), \quad (x_1 + x_2 \leq 1), \quad (x_1 \geq x_2) \right\}$$

$$\mu_3 = \left\{ (0 \leq x_1), (0 \leq x_2), (x_1 \leq 1), \neg(x_1 + x_2 \leq 1), \quad (x_1 \geq x_2) \right\}$$

Volume Computation

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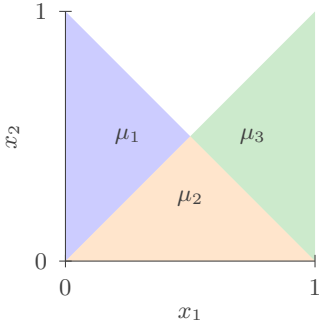

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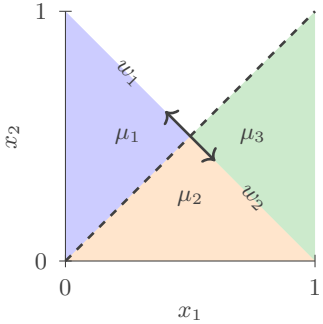
$$\mathsf{vol}(\varphi) \stackrel{\text{def}}{=} \sum_{\mu \ \models \ \varphi} \int_{\mu} 1 \, \mathrm{d}\mathbf{x}$$

$$\Pr\left[\ (x_1 \geq x_2) \ \middle| \ \varphi \ \right] = \frac{\mathsf{vol}(\varphi \wedge (x_1 \geq x_2))}{\mathsf{vol}(\varphi)}$$

Weighted Model Integration

Background & State of the Art



$$\varphi \stackrel{\text{def}}{=} (0 \leq x_1) \wedge (0 \leq x_2) \wedge ((x_1 + x_2 \leq 1) \vee ((x_1 \geq x_2) \wedge (x_1 \leq 1)))$$

$$w \stackrel{\text{def}}{=} ite\big((x_1 \geq x_2), w_1 \stackrel{\text{def}}{=} x_1^2, w_2 \stackrel{\text{def}}{=} x_1 x_2\big)$$

$$\mu_1 = \left\{ \ (0 \leq x_1), (0 \leq x_2), (x_1 \leq 1), \quad (x_1 + x_2 \leq 1), \neg(x_1 \geq x_2) \ \right\}$$

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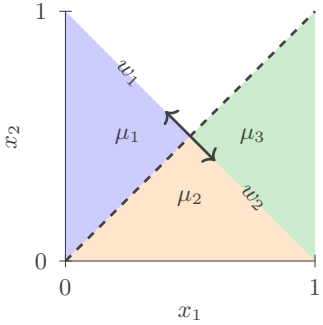
Weighted Model Integration

Background & State of the Art



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$$\mathsf{WMI}(\varphi, w|\mathbf{x}) \stackrel{\text{def}}{=} \sum_{\mu \ \models \ \varphi} \int_{\mu} w(\mathbf{x}) \, \mathrm{d}\mathbf{x}$$

$$\Pr\left[\ (x_1 \geq x_2) \ \middle| \ \varphi \ \right] = \frac{\mathsf{WMI}(\varphi \wedge (x_1 \geq x_2), w|\mathbf{x})}{\mathsf{WMI}(\varphi, w|\mathbf{x})}$$

State-of-the-art WMI Solvers

Background & State of the Art

SMT-based WMI Solver: SAE4WMI[1]

- Use **projected AllSMT** to enumerate convex bodies.
- By augmenting the input, makes the enumeration aware of the structure of the weight function.
- The weight function is **Feasibly Integrable** in each body, enabling direct computation.

KC-based WMI Solvers: XADD[2]

- Compile problem into an algebraic circuit.
- Perform the weighted integration by "resolving" one variable at a time.

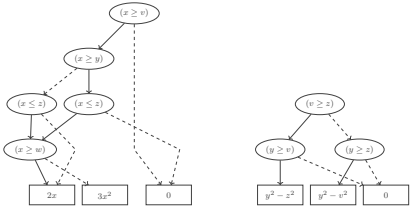

Figure: XADDs. Integrating out x results in the right XADD.

[1] G. Spallitta, G. Masina, P. Morettin, A. Passerini, and R. Sebastiani. "Enhancing SMT-based Weighted Model Integration by structure awareness". In: *AI* (2024)

[2] S. Kolb, M. Mladenov, S. Sanner, V. Belle, and K. Kersting. "Efficient Symbolic Integration for Probabilistic Inference". In: *IJCAI*. 2018

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Approach #1: $\mathcal{T}$-reduced query

Novel Contributions / KCMT for Volume Computation

$\mathcal{T}$-reduced φ D4

INPUT $\varphi, \mathbf{x}, \mathbf{A}$

$\tau = \bigwedge \mathcal{T}\text{-lemmas}\left(\varphi\right)$ (*TDDNNF*)

$\widehat{\varphi} = (\varphi \wedge \tau)\left[\mathbf{x}/\mathbf{B}\right]$

$\widehat{\phi} = \texttt{d-DNNF}\left(\widehat{\varphi}\right)$ (*D4*)

for $\mu \in \mathsf{TA}\left(\widehat{\phi}\right)$ **do** (*DECDNNF*)

  yield $\mu = \mu\left[\mathbf{B}/\mathbf{x}\right]$

end

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Novel Contributions / KCMT for Volume Computation

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end

$\mathcal{T}$-reduced φ PYSDD

INPUT $\varphi, \mathbf{x}, \mathbf{A}$

$\tau = \bigwedge \mathcal{T}\text{-lemmas}\left(\varphi\right)$ (*TDDNNF*)

$\widehat{\varphi} = (\varphi \wedge \tau)\left[\mathbf{x}/\mathbf{B}\right]$

$\widehat{\phi} = \texttt{SDD}\left(\widehat{\varphi}\right)$ (*PYSDD*)

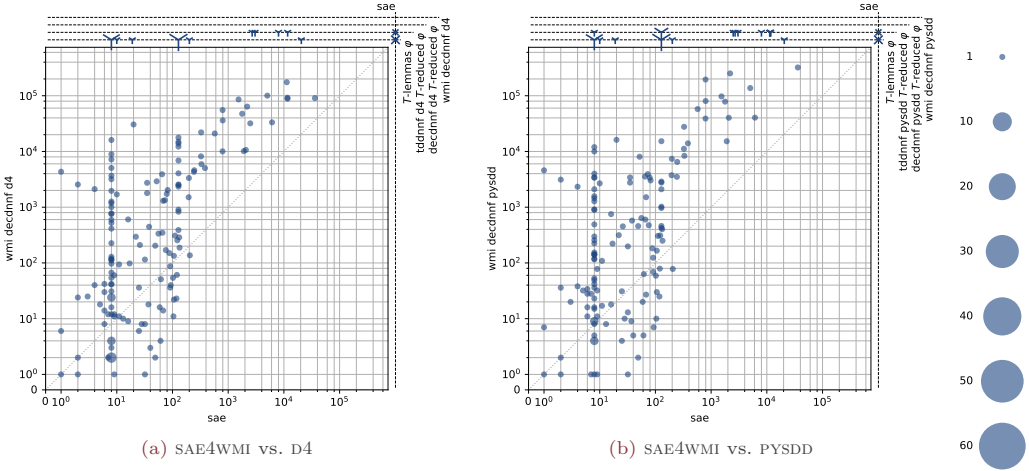
for $\mu \in \mathsf{TA}\left(\widehat{\phi}\right)$ **do** (*DECDNNF*)

  yield $\mu = \mu\left[\mathbf{B}/\mathbf{x}\right]$

end

Approach #1: $\mathcal{T}$-reduced query

Novel Contributions / KCMT for Volume Computation



(a) SAE4WMI vs. D4

(b) SAE4WMI vs. PYSDD

Figure: Number of polytopes.

Approach #2: Existentially-quantify out and condition

Novel Contributions / KCMT for Volume Computation

$\exists \mathbf{B}$ D4

INPUT $\varphi, \mathbf{x}, \mathbf{A}$

$\tau = \bigwedge \mathcal{T}\text{-lemmas}\left(\varphi\right)$ (*TDDNNF*)

$\widehat{\varphi} = (\varphi \wedge \tau)\left[\mathbf{x}/\mathbf{B}\right]$

$\widehat{\phi} = \texttt{d-DNNF}\left(\widehat{\varphi}\right)$ (*D4*)

$\phi^{\mathbf{A}} = \texttt{d-DNNF}\left(\exists \mathbf{B}.\widehat{\varphi}\right)$ (*D4*)

for $\mu^{\mathbf{A}} \in \mathsf{TA}\left(\phi^{\mathbf{A}}\right)$ **do** (*DECDNNF*)

  for $\mu \in \mathsf{TA}\left(\widehat{\phi}_{\left[\mu^{\mathbf{A}}\right]}\right)$ **do** (*DECDNNF*)

    yield $\mu = \mu^{\mathbf{A}} \wedge (\mu\left[\mathbf{B}/\mathbf{x}\right])$

  end

end

Approach #2: Existentially-quantify out and condition

Novel Contributions / KCMT for Volume Computation

$\exists\mathbf{B}$ D4

INPUT $\varphi, \mathbf{x}, \mathbf{A}$

$\tau = \bigwedge \mathcal{T}\text{-lemmas}\left(\varphi\right)$ (*TDDNNF*)

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for $\mu^{\mathbf{A}} \in \mathsf{TA}\left(\phi^{\mathbf{A}}\right)$ **do** (*DECDNNF*)

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    yield $\mu = \mu^{\mathbf{A}} \wedge (\mu\left[\mathbf{B}/\mathbf{x}\right])$

  end

end

$\exists\mathbf{B}$ PYSDD

INPUT $\varphi, \mathbf{x}, \mathbf{A}$

$\tau = \bigwedge \mathcal{T}\text{-lemmas}\left(\varphi\right)$ (*TDDNNF*)

$\widehat{\varphi} = (\varphi \wedge \tau)\left[\mathbf{x}/\mathbf{B}\right]$

$\widehat{\phi} = \texttt{SDD}\left(\widehat{\varphi}\right)$ (*PYSDD*)

$\phi^{\mathbf{A}} = \exists\mathbf{B}.\widehat{\phi}$ (*PYSDD*)

for $\mu^{\mathbf{A}} \in \mathsf{TA}\left(\phi^{\mathbf{A}}\right)$ **do** (*DECDNNF*)

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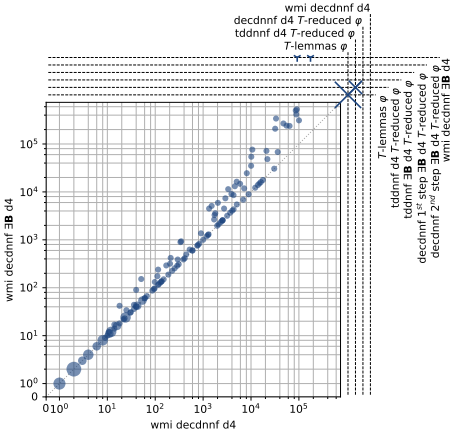
    yield $\mu = \mu^{\mathbf{A}} \wedge (\mu\left[\mathbf{B}/\mathbf{x}\right])$

  end

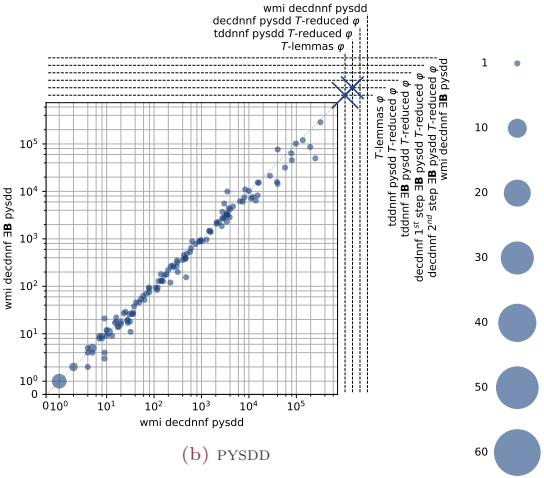
end

Approach #2: Existentially-quantify out and condition

Novel Contributions / KCMT for Volume Computation



(a) D4



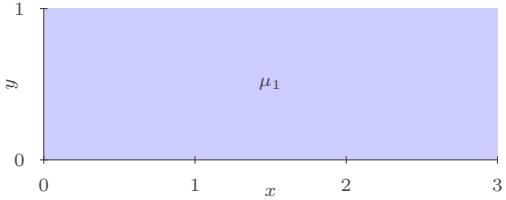
(b) PYSDD

Figure: Number of polytopes.

$\mathcal{T}$-reduction under scrutiny

Novel Contributions / KCMT for Volume Computation

$$\varphi \stackrel{\text{def}}{=} (y \geq 0) \wedge (y \leq 1) \wedge (x \geq 0) \wedge (x \leq 3) \wedge ((x \leq 4) \vee (x \leq 1) \vee (x \geq 2))$$



$\mathcal{T}$-reduction under scrutiny

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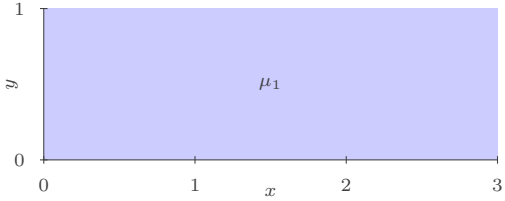
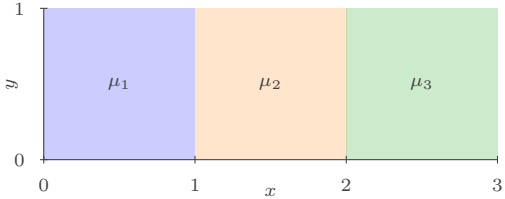


$$\widehat{\varphi} = \varphi \wedge (\overbrace{\neg(x \leq 1) \vee \neg(x \geq 2)}^{\mathcal{T}\text{-lemmas}(\varphi)})$$



Approach #3: Filtering a posteriori

Novel Contributions / KCMT for Volume Computation

Filtering a posteriori D4

INPUT $\varphi, \mathbf{x}, \mathbf{A}$

$\phi = \texttt{d-DNNF}\left(\varphi\left[\mathbf{x}/\mathbf{B}\right]\right)$ (*D4*)

for $\mu \in \mathsf{TA}\left(\phi\right)$ **do** (*DECDNNF*)

  $\mu^{\mathbf{A}} \wedge \mu^{\mathcal{LRA}} = \mu\left[\mathbf{B}/\mathbf{x}\right]$

  if $\mathcal{T}$-$\mathsf{SAT}\left(\mu^{\mathcal{LRA}}\right)$ **then** (*MATHSAT*)

    yield $\mu = \mu^{\mathbf{A}} \wedge \mu^{\mathcal{LRA}}$

  end

end

Approach #3: Filtering a posteriori

Novel Contributions / KCMT for Volume Computation

Filtering a posteriori D4

INPUT $\varphi, \mathbf{x}, \mathbf{A}$

$\phi = \texttt{d-DNNF}\left(\varphi\left[\mathbf{x}/\mathbf{B}\right]\right)$ (*D4*)

for $\mu \in \mathsf{TA}\left(\phi\right)$ **do** (*DECDNNF*)

  $\mu^{\mathbf{A}} \wedge \mu^{\mathcal{LRA}} = \mu\left[\mathbf{B}/\mathbf{x}\right]$

  if $\mathcal{T}$-$\mathsf{SAT}\left(\mu^{\mathcal{LRA}}\right)$ **then** (*MATHSAT*)

    yield $\mu = \mu^{\mathbf{A}} \wedge \mu^{\mathcal{LRA}}$

  end

end

Filtering a posteriori PYSDD

INPUT $\varphi, \mathbf{x}, \mathbf{A}$

$\phi = \texttt{SDD}\left(\varphi\left[\mathbf{x}/\mathbf{B}\right]\right)$ (*PYSDD*)

for $\mu \in \mathsf{TA}\left(\phi\right)$ **do** (*DECDNNF*)

  $\mu^{\mathbf{A}} \wedge \mu^{\mathcal{LRA}} = \mu\left[\mathbf{B}/\mathbf{x}\right]$

  if $\mathcal{T}$-$\mathsf{SAT}\left(\mu^{\mathcal{LRA}}\right)$ **then** (*MATHSAT*)

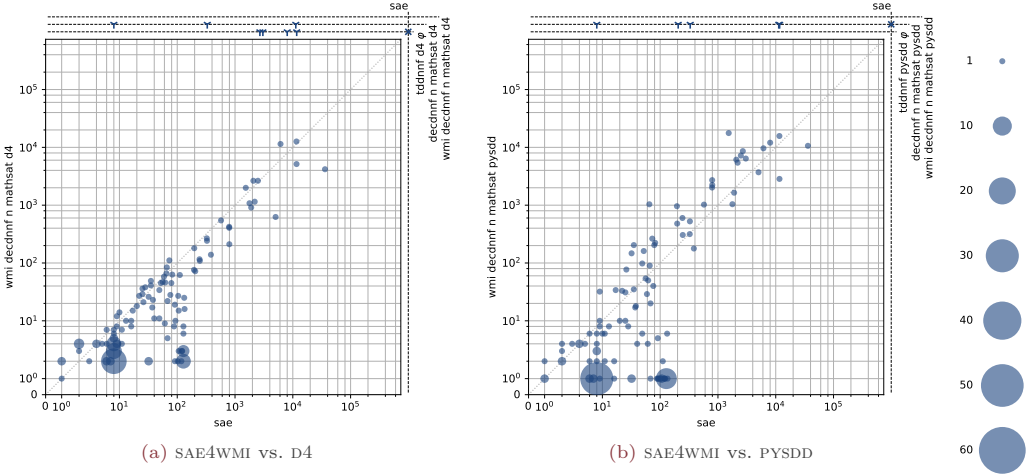
    yield $\mu = \mu^{\mathbf{A}} \wedge \mu^{\mathcal{LRA}}$

  end

end

Approach #3: Filtering a posteriori

Novel Contributions / KCMT for Volume Computation



(a) SAE4WMI vs. D4

(b) SAE4WMI vs. PYSDD

Figure: Number of polytopes.

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Approach #4: $\mathcal{T}$-reduced structure-aware query

Novel Contributions / KCMT for WMI

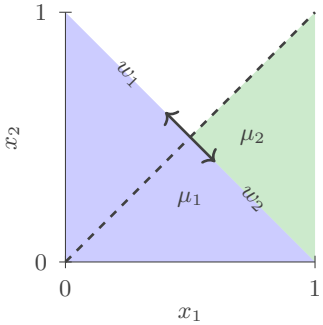

Figure: Enumerating from φ

$$\varphi \stackrel{\text{def}}{=} (0 \leq x_1) \wedge (0 \leq x_2) \wedge ((x_1 + x_2 \leq 1) \vee ((x_1 \geq x_2) \wedge (x_1 \leq 1)))$$

$$w \stackrel{\text{def}}{=} ite\big((x_1 \geq x_2), w_1 \stackrel{\text{def}}{=} x_1^2, w_2 \stackrel{\text{def}}{=} x_1 x_2\big)$$

[1] G. Spallitta, G. Masina, P. Morettin, A. Passerini, and R. Sebastiani. "Enhancing SMT-based Weighted Model Integration by structure awareness". In: *AI* (2024)

Approach #4: $\mathcal{T}$-reduced structure-aware query

Novel Contributions / KCMT for WMI

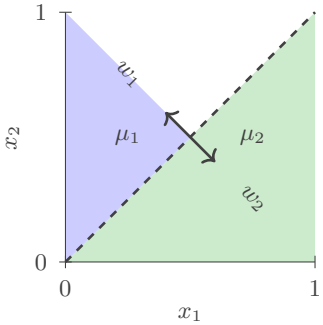

Figure: Enumerating from $\varphi \wedge \mathsf{skeleton}(w)$

$$\varphi \stackrel{\text{def}}{=} (0 \leq x_1) \wedge (0 \leq x_2) \wedge ((x_1 + x_2 \leq 1) \vee ((x_1 \geq x_2) \wedge (x_1 \leq 1)))$$

$$w \stackrel{\text{def}}{=} ite\big((x_1 \geq x_2), w_1 \stackrel{\text{def}}{=} x_1^2, w_2 \stackrel{\text{def}}{=} x_1 x_2\big)$$

Conditional skeleton[1]

- $\mathsf{skeleton}(w)$ is valid
- any partial assignment satisfying $\mathsf{skeleton}(w)$ identifies an unconditional weight function

$$\mathsf{skeleton}(w) = (x_1 \geq x_2) \vee \neg(x_1 \geq x_2)$$

[1] G. Spallitta, G. Masina, P. Morettin, A. Passerini, and R. Sebastiani. "Enhancing SMT-based Weighted Model Integration by structure awareness". In: *AI* (2024)

Approach #4: $\mathcal{T}$-reduced structure-aware query

Novel Contributions / KCMT for WMI

$\mathcal{T}$-reduced φ^{sk}

INPUT $\varphi, w, \mathbf{x}, \mathbf{A}$

$\varphi^{\mathsf{sk}} = \varphi \wedge \mathsf{skeleton}(w)$

$\tau = \bigwedge \mathcal{T}\text{-}\mathsf{lemmas}\left(\varphi^{\mathsf{sk}}\right)$ (*TDDNNF*)

$\widehat{\phi}^{\mathsf{sk}} = \mathtt{d\text{-}DNNF}\left((\varphi^{\mathsf{sk}} \wedge \tau)\left[\mathbf{x}/\mathbf{B}\right]\right)$ (*D4*)

for $\mu \in \mathsf{TA}\left(\widehat{\phi}^{\mathsf{sk}}\right)$ **do** (*DECDNNF*)

  yield $\mu = \mu\left[\mathbf{B}/\mathbf{x}\right]$

end

Approach #4: $\mathcal{T}$-reduced structure-aware query

Novel Contributions / KCMT for WMI

$\mathcal{T}$-reduced φ^{sk}

INPUT $\varphi, w, \mathbf{x}, \mathbf{A}$

1 $\varphi^{\mathsf{sk}} = \varphi \wedge \mathsf{skeleton}(w)$

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4 **for** $\mu \in \mathsf{TA}\left(\widehat{\phi}^{\mathsf{sk}}\right)$ **do** (*DECDNNF*)

5 **yield** $\mu = \mu\left[\mathbf{B}/\mathbf{x}\right]$

6 **end**

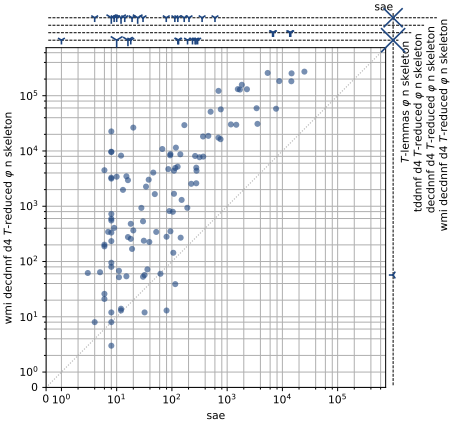

Figure: Number of polytopes.

Approach #5: Filtering a posteriori structure-aware query

Novel Contributions / KCMT for WMI

Filtering a posteriori φ^{sk}

INPUT $\varphi, w, \mathbf{x}, \mathbf{A}$

$\phi^{\mathsf{sk}} = \mathsf{d\text{-}DNNF}\Big((\varphi \wedge \mathsf{skeleton}(w))\Big[\mathbf{x}\big/\mathbf{B}\Big]\Big)$ (*D4*)

for $\mu \in \mathsf{TA}\Big(\phi^{\mathsf{sk}}\Big)$ **do** (*DECDNNF*)

  $\mu^{\mathbf{A}} \wedge \mu^{\mathcal{LRA}} = \mu\Big[\mathbf{B}\big/\mathbf{x}\Big]$

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Novel Contributions / KCMT for WMI

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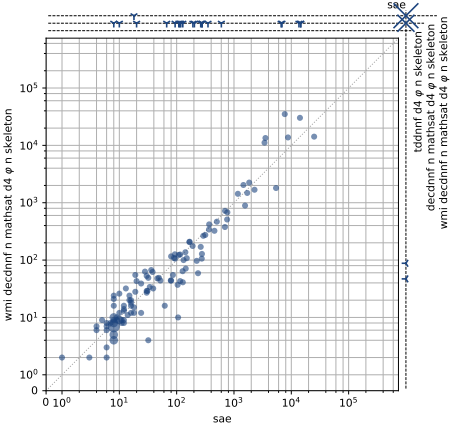
    yield $\mu = \mu^{\mathbf{A}} \wedge \mu^{\mathcal{LRA}}$

  end

end



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Approach #5: Filtering a posteriori structure-aware query

Novel Contributions / KCMT for WMI

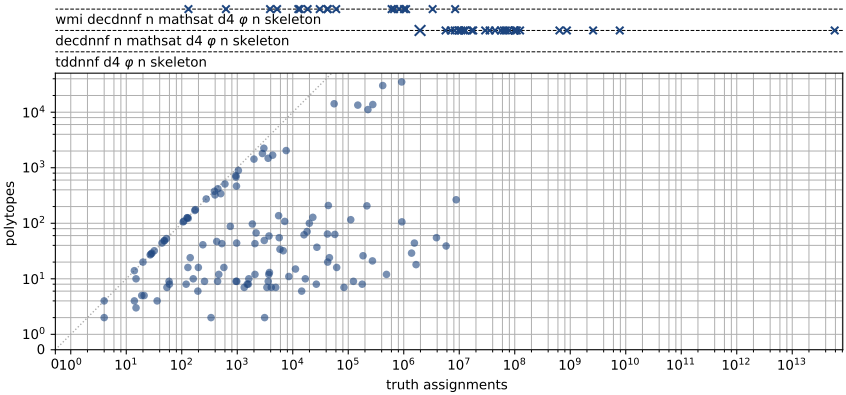

Figure: Number of truth assignments before vs. after MATHSAT filtering.

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Analysis
Conclusions

- $\mathcal{T}$-reduction is counterproductive for short partial-assignment enumeration
  - **Overly strict**: it rules out *all* $\mathcal{T}$-inconsistent truth assignments.
  - **Prevents compactness**: forbids shorter partial assignments that allows $\mathcal{T}$-inconsistent extensions.
  - **Fragmentation**: results in a larger number of polytopes compared to SAE4WMI or a-posteriori filtering.

[1] G. S. Tseitin. "On the Complexity of Derivation in Propositional Calculus". In: *Automation of Reasoning: 2: Classical Papers on Computational Logic 1967–1970*. 1983

[2] G. Masina, G. Spallitta, and R. Sebastiani. "On CNF Conversion for SAT and SMT Enumeration". In: *Journal of AI Research* (2025)

[3] V. Derkinderen and J.-M. Lagniez. "Circuit-aware d-DNNF compilation". In: *IJCAI*. 2025

[4] A. Darwiche. "SDD: A New Canonical Representation of Propositional Knowledge Base". In: *IJCAI*. 2011

[5] R. Sebastiani. "Entailment vs. Verification for Partial-Assignment Satisfiability and Enumeration". In: *CADE*. 2025

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  - **Overly strict**: it rules out *all* $\mathcal{T}$-inconsistent truth assignments.
  - **Prevents compactness**: forbids shorter partial assignments that allows $\mathcal{T}$-inconsistent extensions.
  - **Fragmentation**: results in a larger number of polytopes compared to SAE4WMI or a-posteriori filtering.
- Why a-posteriori filtering works so well
  - **Avoids `CNF`-ization**: Tseitin transformation[1] inflates partial assignments[2], whereas bottom-up (PYSDD) and clever top-down (D4) compilation[3] produce shorter ones.
  - **Canonicity of `SDD`s**[4]: Enables *entailment-based* assignments rather than just *verification-based*, yielding more concise output[5].

[1] G. S. Tseitin. "On the Complexity of Derivation in Propositional Calculus". In: *Automation of Reasoning: 2: Classical Papers on Computational Logic 1967–1970*. 1983

[2] G. Masina, G. Spallitta, and R. Sebastiani. "On CNF Conversion for SAT and SMT Enumeration". In: *Journal of AI Research* (2025)

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Future Work

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- Early-pruning during compilation
  - "Weaker" $\mathcal{T}$-reduction technique

Q&A

Thank you for the attention